Navigation Guide

How to run it, and what every screen is for — Labor Forecasting & Workforce Intelligence

How to get in, what to do first, and what every screen is for.

This application forecasts **a file you upload**. There is no built-in dataset, no sample to fall back on, and nothing is kept after a restart. If a page is empty, it is because nothing has been uploaded yet.

For what the file should contain, see the companion Input Data Template guide.

Contents

1 Running it
2 Signing in
3 The first five minutes
4 The sidebar
5 Data Quality
6 Labor Forecasting — the eight tabs
7 AI Recommendations
8 AI Assistant
9 Reading conventions
10 Say this out loud in a demo
11 When something looks wrong

Running it

Two processes.

```
cd backend_sodexo && ../.venv/Scripts/python.exe -m uvicorn app.main:app --host 127.0.0.1 --p
ort 8300 --reload

cd frontend_sodexo && npm run dev
```

Then open **http://localhost:5183**.

Use `localhost`, not `127.0.0.1` — Vite binds IPv6 loopback, so the IP form will look dead when the server is fine.

Configuration

Everything lives in `.env.sodexo` (gitignored). See `.env.example` for the full list; these two change behaviour most:

Setting	Effect
`RESET_ON_STARTUP`	Default **true** — wipes all data on every startup. Set `false` for local work.
`OPENAI_API_KEY`	Enables AI column mapping. Without it, mapping falls back to detection from names and types.

Signing in

Nothing is reachable without a session.

Username	Password	Role
`admin`	`Admin@2026`	Administrator

Username	Password	Role
`manager`	`Manager@2026`	Operations Manager
`analyst`	`Analyst@2026`	Data Analyst

The role shows in the header but **gates nothing** — all three see the same thing. Say that plainly rather than implying access control exists.

Sessions last 8 hours. Change these before any deployment that is not a local demo.

The first five minutes

The app starts empty, every time. On startup it deletes all data: the uploaded datasets, the files behind them, and every trained model. This is a single-tenant POC, and a page still holding the last person's numbers reads as your data until someone checks the series name.

#	Where	Action
1	**Data Quality** → **Upload Data**	Drop your CSV or Excel file
2	same panel	Check the proposed mapping, then **Save mapping**
3	dataset row	**Train**
4	**Labor Forecasting**	Everything is now live

No file to hand? **Download example file** in the upload panel gives you a working one — 1,095 rows across three sites, with a planned-capacity column and three drivers, so every feature has something to show.

Training runs the full 13-model portfolio per series. A single series takes roughly two minutes.

The sidebar

```
Data Quality                ← opens here; upload lives here
Labor Forecasting           ← eight tabs, in workflow order
    Demand Drivers & Feature Mapping
    Data Analysis
    Data Pre-processing
    Model Training
    Forecast Explorer
    Drift Monitoring
    Staffing Impact
    Model Performance
AI Recommendations
AI Assistant
```

The eight tabs read top to bottom as the actual workflow: what goes in → what the data looks like → what the models see → training → the forecast → can I still trust it → what it means for the roster → how good was it.

Most tabs have a **series picker**. Every series in it comes from an uploaded file; a dataset appears once trained, never as a dead option that errors when selected.

Data Quality

Two things: the upload panel, and the health of what you uploaded.

Upload Data

Drop a file → the columns are read and a mapping proposed → confirm it →

Save mapping → **Train**.

What can I upload? expands to the accepted file types, size and row limits, minimum history, and what each column role is for — read from the server, so it always matches what the code enforces.

Field	What to put in it
Date column	The time axis. Columns that parse as dates are marked ✓.
Forecast	The numeric column to predict.
Planned / capacity	Optional. What was already rostered or budgeted. Unlocks Staffing Impact.
Group by	Optional. One series per value, up to 40.
Frequency	Hourly / Daily / Weekly / Monthly. Inferred from your dates.
Combine duplicates by	Sum or Average, when rows share a period.
Horizon	Periods to forecast ahead.
Drivers	Numeric columns fed to the models as inputs.

The panel says whether the mapping came from **AI** or from **names and types**, and shows the model's one-line reasoning when it was AI. Either way you confirm before anything trains.

Save mapping validates immediately — a mapping that yields no usable rows is reported there, not after a training run.

The health table

Per file: rows read, how many are **usable**, how many are not, how many series it produces, and a quality grade. Problems are listed underneath in plain language — unreadable dates, non-numeric targets, periods missing a planned value, duplicate periods.

Unusable means a row the mapping cannot use: no readable date, or no number in the target column. Those rows are dropped before training, so counting them as loaded would overstate what the models saw.

Labor Forecasting — the eight tabs

Demand Drivers & Feature Mapping

The target and every input column, per series. **Columns are clickable** — switching one off persists and changes what the next training run uses.

Column choices belong to the file, so a file split into groups shares one mapping across all its series. The panel says so.

Data Analysis

Historical trend with a smoothed line over it, demand by hour of day, by day of week, by season, and the on/off lift of any driver that is effectively a flag. Every chart carries a one-line finding computed from the same data.

Data Pre-processing

What happens between your file and training: history length, resolved seasonality, forecast horizon, the validation plan, and **a preview of the first 10 rows actually fed to the models**, with the target last.

Model Training

Train All Models runs the full portfolio. While it runs: a live badge, a progress bar (`n / total`), and a feed of models as they finish.

The leaderboard shows every model with status, runtime and metrics. A model that cannot run is marked `not_applicable` **with a reason** — never silently dropped.

By default **every model runs**. `TRAINING_PROFILE=fast` skips the three slowest (LSTM and both Auto ARIMA variants), worth it only when a large multi-series upload would otherwise take hours.

Forecast Explorer

Line	Meaning
Grey, solid	Actual — what really happened
Orange, solid	Backtest — what the model predicted for periods already known
Green, dashed	Future forecast. Dashed because nothing here is verified
Vertical dashed	Where known ends and predicted begins

How to read it honestly: compare orange against grey in the overlap. If orange is flatter — hugging the middle while grey swings — the model is predicting near the average rather than tracking the pattern, and a high accuracy number is coming from a stable series rather than a skilful model.

Drift Monitoring

Answers "can I still trust this model?" Compares the recent demand signal against the previous period, and the shape of the day now against before.

Thresholds: within 15% **Stable**, 15–30% **Watch**, beyond 30% **Drift Detected** (retrain).

The header states how strongly the signal actually tracks the target, and warns when that link is weak rather than asserting it is *the* driver.

Staffing Impact

Required vs planned, with the gap colour-coded: red understaffed, gold overstaffed, green balanced. A small surplus counts as balanced — planning exactly to forecast leaves no slack for one absence.

Needs a **planned-capacity column**. Without one the tab says so and tells you how to add it. Periods where the plan has no value are left out rather than counted as zero planned.

Model Performance

MAPE by model, champion highlighted, on a 0-based axis. MAPE rather than accuracy because accuracy values cluster in the 90s and every bar looks identical; on MAPE the champion is visibly the shortest bar.

AI Recommendations

Filter bar, KPI tiles, clickable visuals, then the cards. All filters are built from the data, so they show your own group names.

Each card answers four questions — what's happening, what to do, why it matters, and the figures it fired on. Where another series has spare capacity at the same moment, the action says to move it and names the source.

These are **rule-based, not an optimiser**, and they need a planned-capacity column. No column, no cards.

No cost or savings figures: an uploaded file has no pay rates, and a guessed number is worse than none.

AI Assistant

Chat over the data you uploaded. Uses OpenAI when a key is configured, and falls back to a template summariser over the same retrieved facts when not — grounded either way.

- **On-topic only.** Off-topic questions are declined before any retrieval.
- **Scoped by your own names.** The vocabulary comes from your series labels, so "how is harbour site doing" narrows to that series. Words common to every series are ignored — otherwise a file split into "Depot plan / North" and "Depot plan / South" would match everything on "depot".
- **Follow-ups keep context.** "why?", "what about there?" inherit the previous turn's series; a self-contained question resets.
- **Units are yours.** A daily file of labour hours reads as *"Labour Hours Required per day"*, not a borrowed unit.

It answers from the same services the pages use — never raw database access, never invented numbers.

Reading conventions

- **Dashed = not verified.** Any dashed line is beyond the last actual.
- `not_applicable` **always carries a reason.** Nothing is silently dropped.
- **Cross-filtering:** clicking a segment filters the page; a visual never filters itself; clicking the selection again clears it.
- **Empty is honest.** A tab with no data says what is missing and how to supply it, rather than showing a plausible-looking zero.

Say this out loud in a demo

- **All data is the client's own upload.** Nothing ships with the app, and everything is wiped on restart.
- **A high accuracy number is not automatically a good model.** On a stable series, predicting the average scores well. Check the Forecast Explorer overlap before quoting a figure.
- **The mapping is AI-assisted but validated and confirmed.** The model proposes; the code checks every answer against the real columns; the user confirms before training.
- **Drift detection is a percentage threshold**, not a statistical test. Recommendations are rules, not an optimiser.
- **Roles do not gate anything yet.**

When something looks wrong

Symptom	Cause	Fix
Everything is empty	Nothing uploaded, or a restart wiped it	Upload a file, or set `RESET_ON_STARTUP=false`
"No trained model yet"	Dataset mapped but not trained	Data Quality → **Train**
Staffing Impact empty	No planned-capacity column mapped	Map one in Demand Drivers, or add it to the file
No recommendations	Same — they need planned capacity	As above
A model shows `not_applicable`	Read the reason on the row	Usually a driver that is flat at this grain, or too little history
Forecasting daily, expected hourly	Separate date and hour columns	Combine into one timestamp column — see the template doc

Symptom	Cause	Fix
Frontend looks dead on `127.0.0.1`	Vite binds IPv6 only	Use `localhost:5183`
A code change had no effect	Backend served stale code	Restart it; it does not always hot-reload cleanly